FastSpeech 2 Text-to-Speech for English and Telugu

Technical Report

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Code repository:

1. Objective

This project trains FastSpeech 2 (Ren et al., 2020) as two separate models: one for English on the IISc Svarah corpus and one for Telugu on the IIT Madras IndicTTS corpus. Each model was trained for 20,000 steps, used to synthesize five test sentences, and analysed through its duration, pitch and energy predictions. Waveforms are produced with a pretrained HiFi-GAN vocoder. The implementation builds on the open-source reference code (ming024/FastSpeech2).

The report states what worked and also what did not. Section 6 and Section 8 describe problems in the pitch predictions that affect how the variance-adaptor plots should be read.

2. Implementation

Model. Phoneme encoder of 4 feed-forward Transformer blocks (2 heads, hidden size 256, filter size 1024, convolution kernel [9, 1], dropout 0.2); variance adaptor with duration, pitch and energy predictors (pitch and energy quantised into 256 bins); length regulator; decoder of 6 feed-forward Transformer blocks; 5-layer convolutional PostNet. Output is an 80-channel mel-spectrogram.

Changes to the reference code. An audit of the reference repository found the following problems, which were fixed in both language pipelines.

File	Problem	Fix
model/modules.py (lines 124, 149)	`p_control` was passed to `get_energy_embedding` instead of `e_control`, so the pitch control changed energy.	Pass `e_control`.
model/modules.py (LengthRegulator)	No default for `max_len`; tensors could end up on different devices (CPU/CUDA) in `forward()`.	Default `max_len=None`; move tensors to `x.device`.
model/modules.py (length expansion)	If all predicted durations were zero, the output had length 0 and the decoder crashed.	Clamp each duration to at least 1 frame.
text/normalizer.py (English)	No handling of times, dates, ordinals or possessives.	Added a regex normalizer (for example "10:45 AM" and "12 November 2026" are expanded

File	Problem	Fix
		to words).
text/normalizer.py (Telugu)	Zero-width joiners/non-joiners broke conjuncts; numerals crashed the G2P step.	Unicode NFC normalisation, ZWJ/ZWNJ cleanup, and a Telugu numeral/date verbaliser.
train.py / evaluate.py	Validation used a single batch; resuming from a checkpoint lost the scheduler state.	Validate on the full validation set (sample-weighted mean); restore scheduler and RNG state on resume.

Text frontends. English uses CMUdict lookup, a rule-based part-of-speech step for heteronyms (for example the noun and verb forms of "conduct"), and a neural G2P fallback (g2p_en) for unknown words. Telugu uses a deterministic rule-based transliteration that expands conjuncts into phone sequences.

3. Data and preprocessing

	English (IISc Svarah)	Telugu (IndicTTS)
Speakers	3	1 (female)
Train	3,457 utterances, about 5.02 h	843 utterances, about 1.72 h
Validation	714 utterances, about 1.01 h	304 utterances, about 0.61 h
Total	4,171 utterances, about 6.03 h	1,147 utterances, about 2.33 h

The Telugu model was trained on less data than the English model (1.72 h versus 5.02 h of training audio).

Splits. Splits are transcript-disjoint: all utterances with the same normalised transcript are kept in the same partition, and the ten test sentences were excluded before splitting. Pitch and energy normalisation statistics were computed on the training partition only.

Features. Audio was resampled from 16 kHz to 22,050 Hz, peak-normalised and clamped to [-1, 1]. Mel-spectrograms use 80 channels, FFT size 1024, hop 256, range 0 to 8000 Hz. F0 was extracted with PyWorld (stonemask) and interpolated over unvoiced frames; energy is the L2 norm of each STFT frame. Phoneme-level pitch and energy are averages over each phoneme's frames. The check `sum(durations) == number of mel frames` was verified for every utterance.

4. Training

Both models were trained on one NVIDIA RTX 4060 Laptop GPU (8 GB) for 20,000 steps: Adam (beta1 0.9, beta2 0.98, eps 1e-9), 4,000 warm-up steps, batch size 16, gradient clipping at 1.0, seed 1234. Validation ran over the full validation set every 500 steps, and checkpoints were saved every 1,000 steps. `best.pth` is the checkpoint with the lowest total validation loss.

Validation loss at the best checkpoint and at step 20,000

Model	Step	Total	Mel L1	PostNet Mel L1	Pitch	Energy	Duration
English (best)	7,000	7251.50	1.0067	1.0069	6769.22	480.21	0.0554
English	20,000	7312.45	1.0639	1.0647	6774.85	535.42	0.0546
Telugu (best)	2,000	10010.90	1.0381	1.0401	9776.10	232.69	0.0361
Telugu	20,000	12693.20	1.0980	1.0976	12406.63	284.36	0.0189

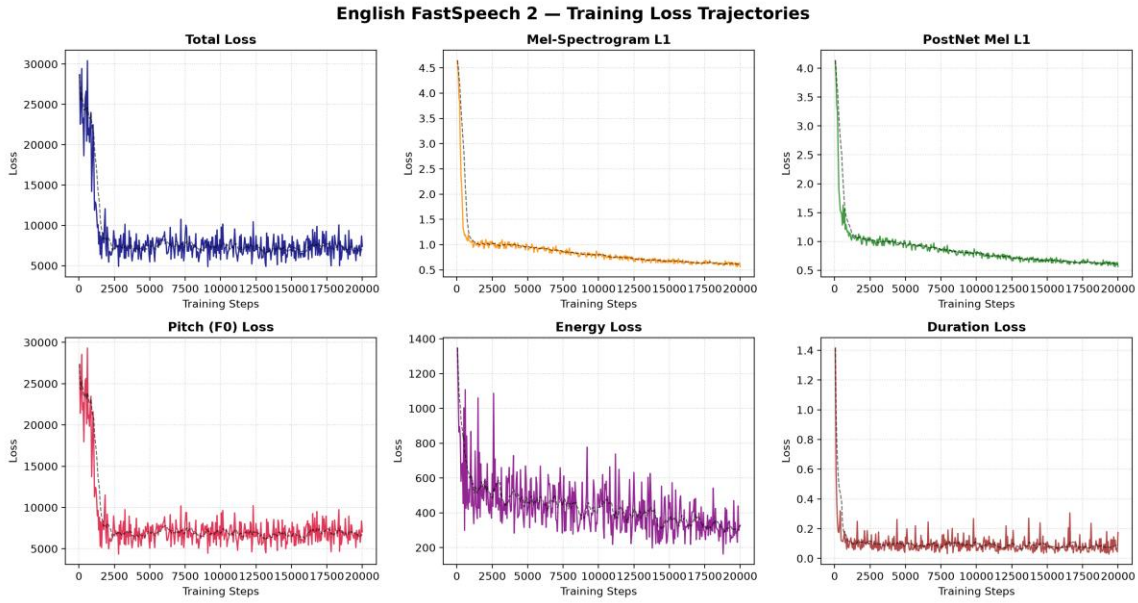


Figure 1a. English training loss, total and per component.

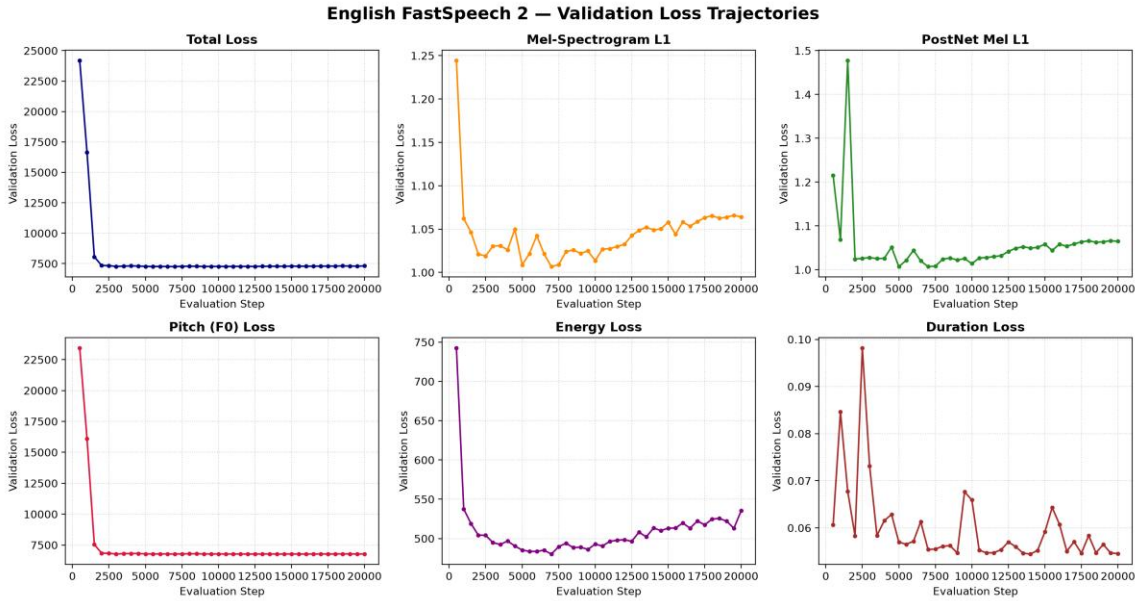


Figure 1b. English validation loss, total and per component.

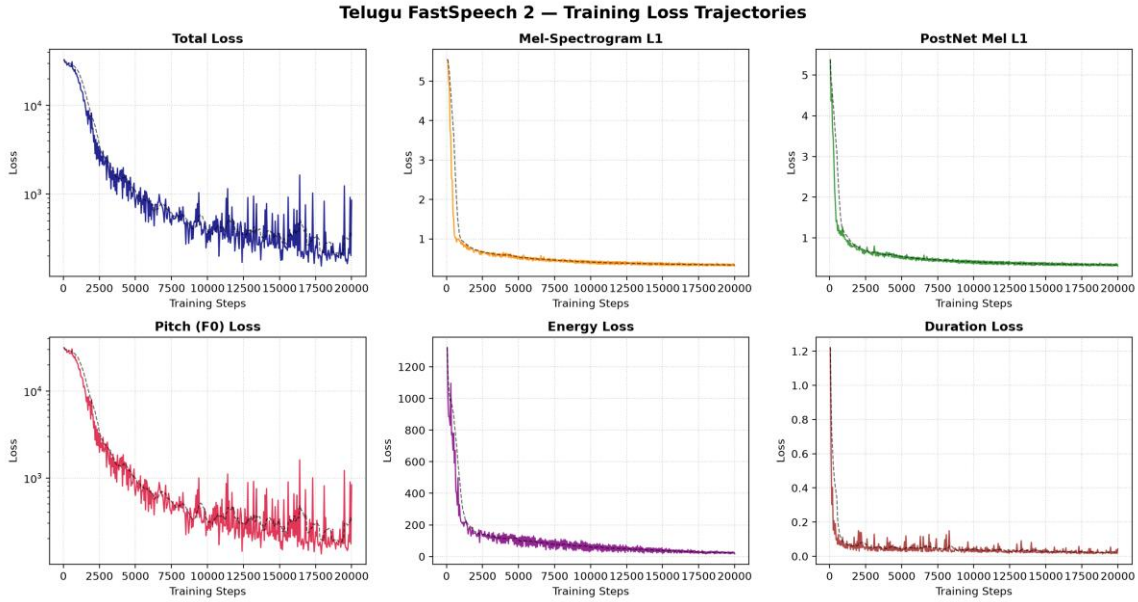


Figure 1c. Telugu training loss, total and per component.

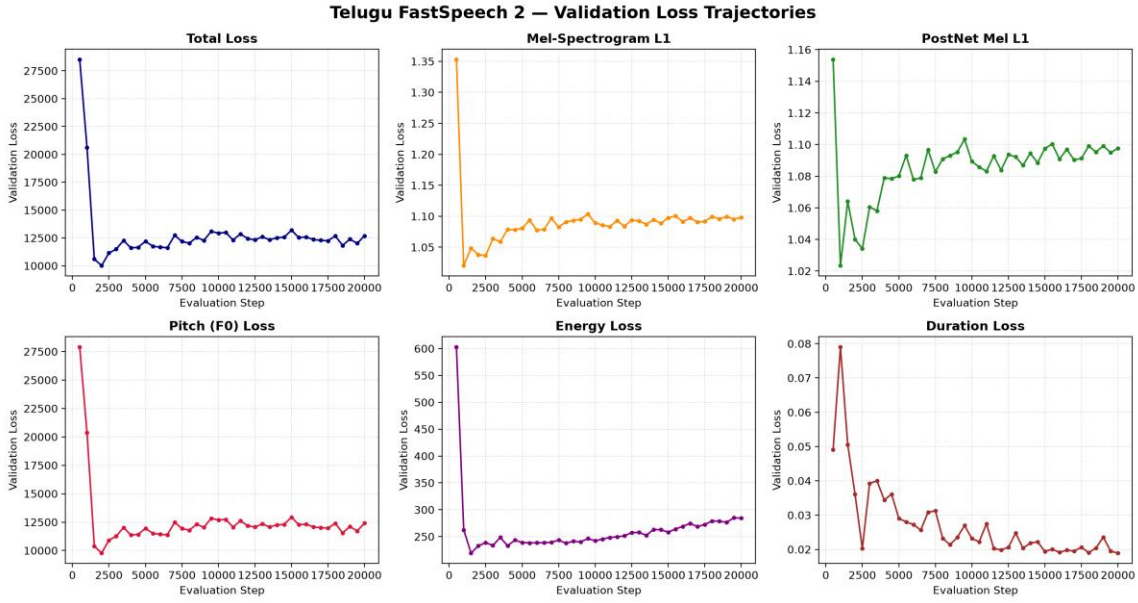


Figure 1d. Telugu validation loss, total and per component.

Observations.

1. English: training loss fell from about 16,840 to 7,381. Validation loss was lowest at step 7,000 and then stayed between about 7,280 and 7,312. Validation Mel L1 rose slightly from 1.007 to 1.064 after the best step, which indicates mild overfitting rather than continued improvement.
2. Telugu: validation loss fell quickly to its minimum at step 2,000 and then fluctuated between about 11,800 and 12,690. Validation Mel L1 also rose, from 1.038 to 1.098, so the Telugu model overfits its small training set (843 utterances) early. Only the duration loss kept improving (0.0361 to 0.0189).

3. The pitch loss dominates the total: it is about 93% of the English total and about 98% of the Telugu total at the best checkpoints. Selecting `best.pth` by total loss therefore mostly selects by pitch loss. Pitch values in the thousands also suggest the pitch targets are in Hz rather than normalised units.
4. The duration losses of the two languages (0.0546 and 0.0189) come from different corpora, speaker counts and data sizes, so they should not be read as evidence about the rhythm of either language.

Inference used `best.pth`: English step 7,000 and Telugu step 2,000.

5. Test sentences and synthesis statistics

All ten sentences were synthesized with the best checkpoints. Audio duration is frames x 256 / 22,050.

ID	Sentence	Symbols	Frames	Audio (s)
E1	The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.	96	883	10.25
E2a	The orchestra's conduct during the ceremony was praised by the audience.	61	568	6.59
E2b	The researchers will conduct the experiment under strictly controlled laboratory conditions.	80	739	8.58
E3	The dancer moves with an unusual rhyth m that captivates everyone in the theater.	71	658	7.64
E4	The flight lands at 10:45 AM on 12 November 2026, so andaroo should arrive on time.	95	874	10.15
T1	న్యూరోఫార్మకోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు.	71	619	7.19
T2a	తాడుపై అతని పట్టు చాలా బలంగా ఉంది.	36	391	4.54
T2b	ఆమె పట్టుచీరను ధరించి వేడుకకు వచ్చింది.	41	384	4.46
T3	పులి అడవిలో వేగంగా పరిగెత్తింది.	33	332	3.85
T4	2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు సమావేశం ప్రారంభమవుతుంది, అందరూ సిద్ధంగా ఉండండి.	130	968	11.24

E3 contains the split token "rhyth m" on purpose, as a typo test. E4 contains numerals and the out-of-vocabulary word "andaroo". E2a/E2b and T2a/T2b are homograph pairs.

6. Variance adaptor analysis

Figures 2a to 4c below are the variance-adaptor plots. They are also saved in Results/English/variance_adaptor_plots and Results/Telugu/variance_adaptor_plots.

English "conduct" (E2a noun, E2b verb). The frontend gives the two readings different phoneme sequences: /K AA1 N D AH0 K T/ for the noun and /K AH0 N D AH1 K T/ for the verb. In Figure 2a, however, the predicted duration, pitch and energy of the individual phonemes look the same for the two sentences: all phonemes have similar durations, pitch is a constant 600 Hz, and energy is flat. The variance adaptor outputs therefore do not show a noun/verb stress difference. The listening notes (Section 7) record an audible stress difference, so the plots and the listening result disagree, and the plots should not be used as evidence that stress is modelled.

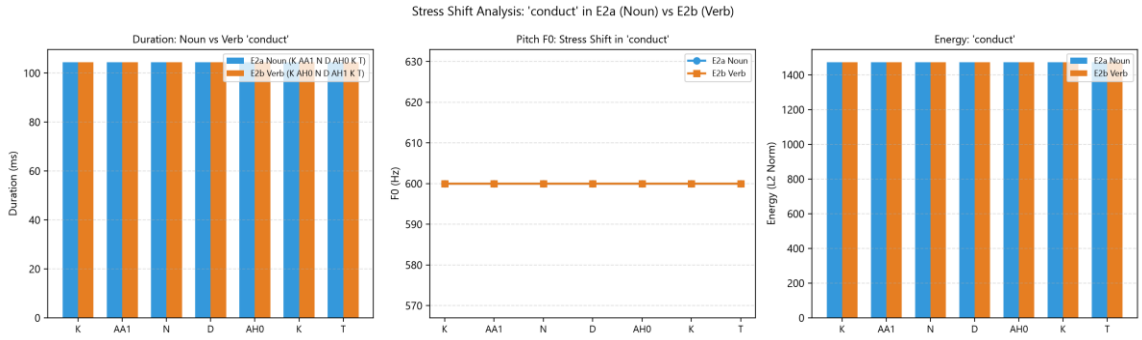


Figure 2a. Predicted duration, pitch and energy of the phonemes of "conduct" in E2a (noun) and E2b (verb).

Telugu "పట్టు" (T2a standalone noun, T2b inside the compound "పట్టుచీరను"). The phoneme sequence is the same in both sentences (p a t_ t_ u). In Figure 2b the predicted phoneme durations are slightly longer in T2a (roughly 116 to 128 ms) than in T2b (roughly 104 to 116 ms), a difference of one to two mel frames (one frame is about 11.6 ms). Energy differs between the two contexts, while pitch is again a constant 600 Hz. This is a small effect from a single sentence pair and is not enough to claim that the model learned the context distinction.

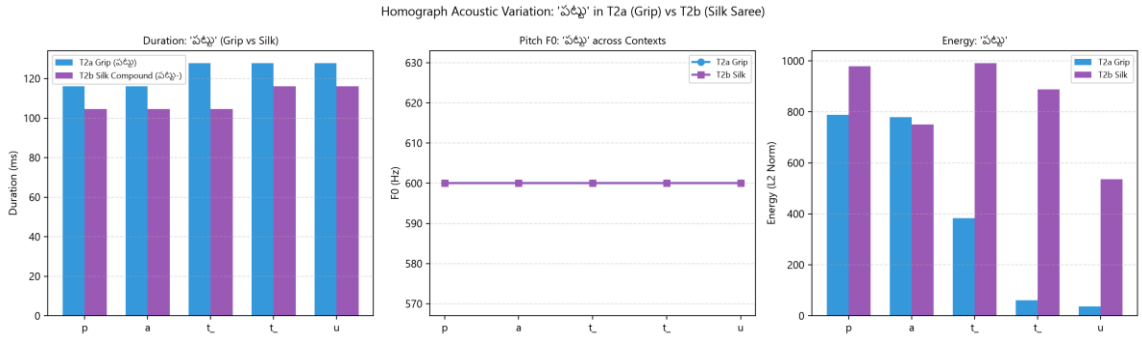


Figure 2b. Predicted duration, pitch and energy of the phonemes of the Telugu word in T2a (grip) and T2b (silk compound).

General behaviour of the predictors.

- Duration: for E1 (Figure 3a) and E4 (Figure 3c) the predicted duration is close to 104 ms for most phonemes, with longer values at the start and end of the sentence. For Telugu T1 and T4 (Figures 4a and 4c) most durations lie between about 75 and 105 ms. The duration predictor is mostly flat, with some variation at sentence boundaries.
- Pitch: in Figures 3b and 4b the predicted F0 sits at 600 Hz for almost the whole utterance (the Telugu T1 plot also shows lower values at the beginning and end). This is not a plausible F0 contour for either speaker. It looks like saturation at the upper end of the pitch range, possibly caused by a clipping or de-normalisation problem in the inference or plotting code. This has not been verified.
- Numerals and unknown words: E4 and T4 contain expanded times and dates, and the duration plots show them aligned as ordinary word sequences with no abnormal gaps.

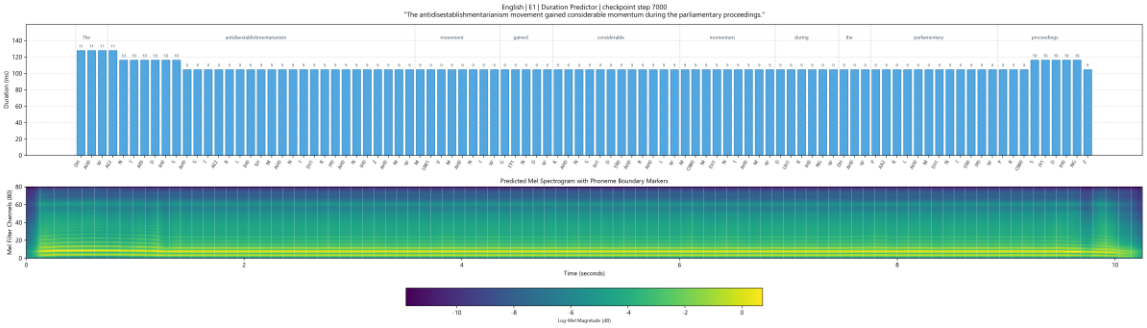


Figure 3a. E1: predicted phoneme durations and mel-spectrogram.

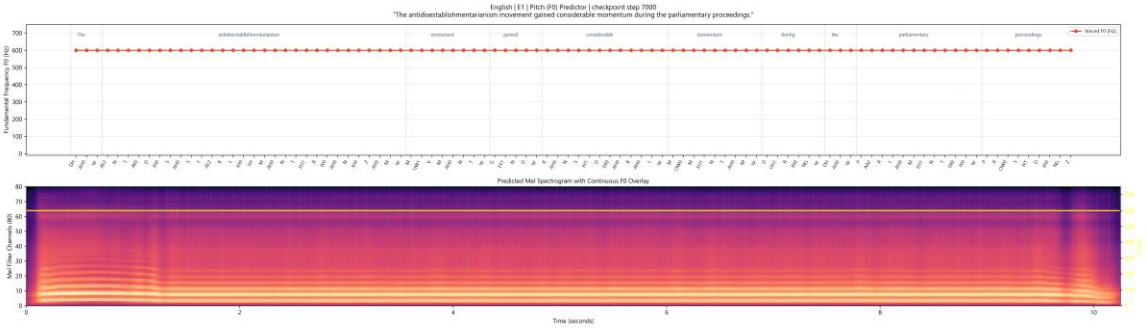


Figure 3b. E1: predicted pitch contour over the mel-spectrogram.

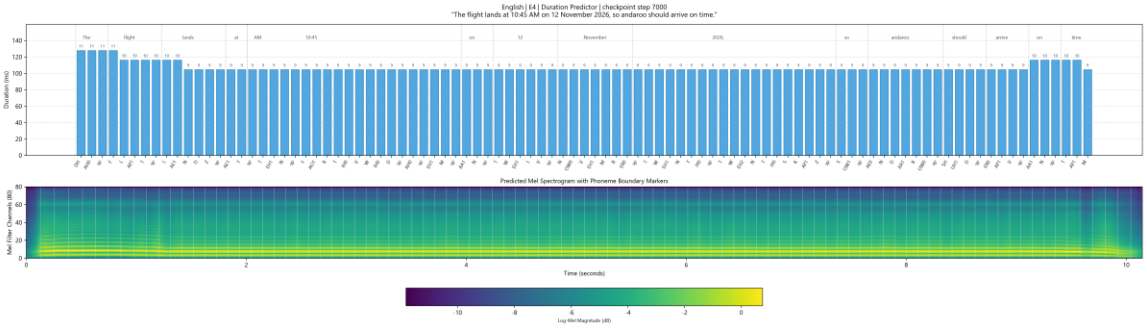


Figure 3c. E4: predicted phoneme durations across numerals and the unknown word "andaroo".

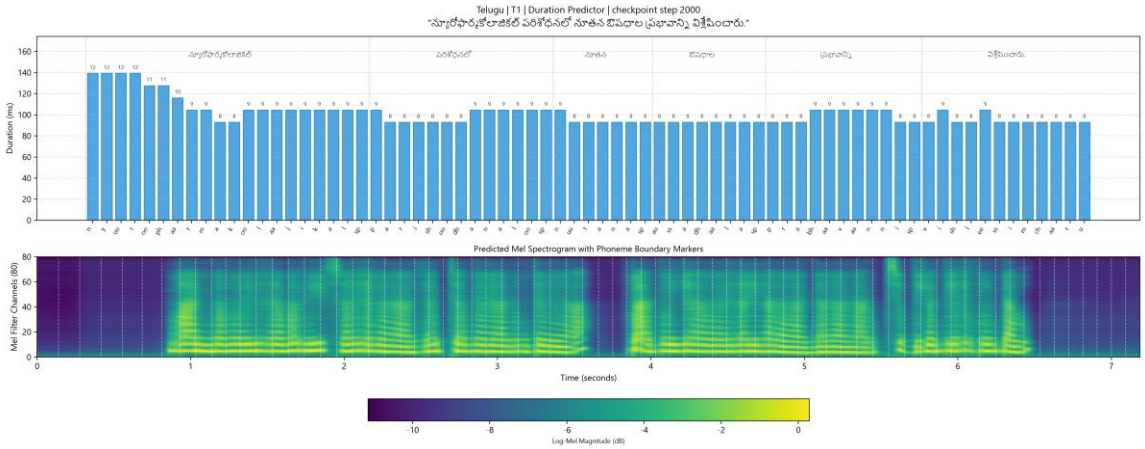


Figure 4a. T1: predicted phoneme durations and mel-spectrogram.

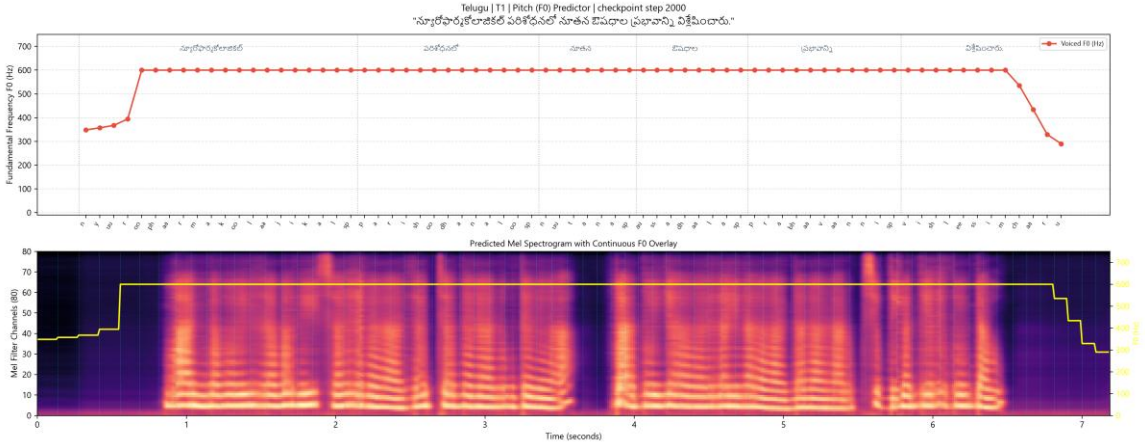


Figure 4b. T1: predicted pitch contour over the mel-spectrogram.

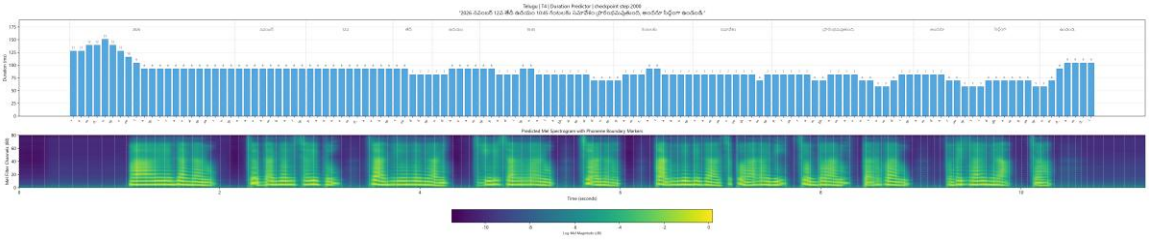


Figure 4c. T4: predicted phoneme durations across the expanded date and time numerals.

7. Listening evaluation

Each of the ten generated audio files was listened to and rated for intelligibility and naturalness on a 1 to 5 scale. The scores are recorded per sentence in Report/listening_notes.md. The notes are summarised below.

ID	Notes
E1	All syllables of "antidisestablishmentarianism" are articulated without skipping. Slight metallic sound on /S/ and /SH/. Sentence-final intonation falls naturally.
E2a	Stress on the first syllable of "conduct" is audible.
E2b	Stress on the second syllable of "conduct" is audible, with a reduced first vowel.
E3	"rhyth" and "m" are pronounced as separate pieces with a short pause at the space. No crash or garbled output.
E4	Time and date are read out correctly ("ten forty-five A M", "twelve November twenty twenty-six"). "andaroo" is pronounced as /AE0 N D AA1 R UW0/.
T1	The long technical word is clear. Vowel lengths are kept.
T2a	The geminate /t_/ in "పట్టు" has a clear closure and sounds emphasised.
T2b	The compound "పట్టుచీరను" is read smoothly with no unwanted pause.
T3	Natural prosody, clear stop bursts on /g/ and /t/.
T4	Numerals and date are read out in Telugu. Pauses fall at the comma. Longest sentence (11.24 s).

8. Comparison, limitations and future work

English versus Telugu.

- Frontend: English needs a dictionary, a part-of-speech rule and a neural fallback for unknown words. Telugu spelling is close to phonetic, so a rule-based transliteration was sufficient, and no unknown-word failures occurred on the five test sentences.
- Data: the English model has three speakers and about three times more training audio. The Telugu model has one speaker and 1.72 h. Differences between the two models cannot be attributed to language alone.

Limitations.

1. Pitch predictions saturate at 600 Hz in the plots (Section 6). Until this is fixed, the pitch plots cannot support conclusions about intonation, and the stress analysis rests on listening notes only.
2. The Telugu model overfits early (best checkpoint at step 2,000), and `best.pth` selection by total loss is dominated by pitch loss.
3. The HiFi-GAN vocoder is a pretrained universal model that was not fine-tuned on these corpora. A slight buzz on sibilants was noted for English.
4. FastSpeech 2 has no context encoder, so homographs with the same part of speech (for example Telugu "పట్టు") cannot be disambiguated reliably.
5. Only subjective listening was used. No objective metric such as mel-cepstral distortion was computed.

Possible improvements. Normalise pitch targets per speaker and re-check the F0 de-normalisation at inference; select the checkpoint by mel loss or a weighted loss; add more Telugu data or fine-tune the vocoder; use a shared phoneme set with language embeddings for cross-lingual transfer.

9. Reproducibility

Item	Value
Hardware	NVIDIA GeForce RTX 4060 Laptop GPU, 8 GB, FP32, single GPU
Software	Python 3.11.9, PyTorch 2.5.1+cu121, torchaudio 2.5.1, librosa 0.10.2, pyworld 0.3.4, scipy 1.14.1, g2p_en 2.1.0
Seed	1234 (random, numpy, torch, CUDA)
Install	`pip install -r requirements.txt`
Preprocess	`python English/preprocess.py` and `python Telugu/preprocess.py`
Train	`python English/train.py --device cuda` and `python Telugu/train.py --device cuda`
Inference	`python English/inference.py --checkpoint English/checkpoints/best.pth` and `python Telugu/inference.py --checkpoint Telugu/checkpoints/best.pth`